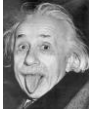


Game Theory



Albert Einstein:
"God does not play dice with the universe."





Stephen Hawking:
"God sometimes throws dice where they can't be seen."





Department of Management Studies
Indian Institute of Technology Madras



Interactive Decision Theory

- Decision Theory
 - You are self-interested and selfish
- Game Theory
 - So is everyone else



"If it's true that we are here to help others, then what exactly are the others here for?"

- George Carlin
(American Standup Comedian: 1937-2008)

The Golden Rule

COMMANDMENT

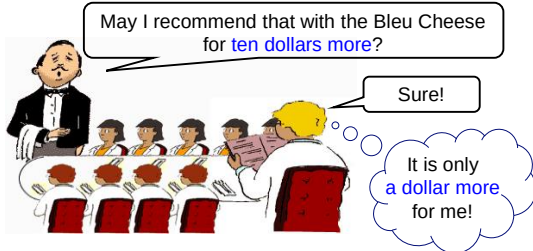
Never assume that your opponents' behavior is fixed.

Predict their reaction to your behavior.

Decision Theory vs. Game Theory

- Ten of you go to a restaurant
- If each of you pays for your own meal...
 - This is a decision problem
- If you all agree to split the bill...
 - Now, this is a game

Restaurant Decision-Making

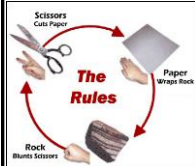


'Check splitting policy' changes incentives.

Understanding Incentives

- ☐ Do seatbelts cause less accidents?
- ☐ Why cannibalize your own sales?
- ☐ Why honor price guarantees to consumers?
- ☐ Why reduce capacity even if it is being used?
- ☐ Does Pakistani Cricket have less punting than Indian Cricket?

Rock-Paper-Scissors GAME



	Player R's Strategies		
	Rock	Scissors	Paper
Player F's Strategies			
Rock	(Draw, Draw)	(Win, Lose)	(Lose, Win)
Scissors	(Lose, Win)	(Draw, Draw)	(Win, Lose)
Paper	(Win, Lose)	(Lose, Win)	(Draw, Draw)

Note: Within the parentheses the first payoff is player F's and the second is for player R. (F's payoff, R's payoff)

CBS & NBC choosing TV Shows (USA):

- They can choose to show a Drama, Comedy, or Sports show.
- Viewers prefer Comedy to Drama, Sports to Comedy, and Drama to Sports, given the choice.
- Neither station knows ahead of time what the other will choose.

Strategy #1: Have a Lot of Territory

The Orange-Throated Lizard:
These males establish large territories, with several females.
The more females the more often they can mate.



Strategy #3: Be Sneaky

The Yellow-Striped-Throated Lizard:
These males are sneaky and can mimic the markings and behavior of females.

Strategy #2: Guard Your Mate

The Blue-Throated Lizard:
These males defend small territories holding just a few females. Because the territories are so small, they can guard their mates carefully.

Toys "R" Us (TRU)



Competitive Landscape

- Toy & department stores
 - Do not compete with TRU on price (50% markups)
- General discounters (Wal-Mart, Target)
 - Cannot compete with TRU on selection (9% markups)
 - Toys exhibit high product differentiation
- Specialized discounters (only TRU)
 - Enjoyed 20% market share nationally
- Warehouse clubs (Costco, Pace)
 - Introduced in the 1980s
 - By 1989, about 200 items in competition

Cause of concern for Toys "R" Us!

Toys "R" Us (continued)

- Market research into the future of warehouse clubs
 - Manufacturers estimated 3-5% annual growth rate
 - Toys "R" Us conducted comprehensive but proprietary study
 - Manufacturers were aware of the study, but not its results
 - Toys "R" Us expected warehouse clubs to grow quite fast
- Idea! Issue statement to manufacturers:
 - "You may not sell to warehouse clubs without losing Toys "R" Us as a customer"
 - Effectively, choose between a \$5 billion market (TRU) and a less than \$500M market (warehouse clubs)

Toys "R" Us: ISSUES (continued)

Executives strived to change the game

- Broker a cooperative agreement among the toy manufacturers
 - Serve as a clearinghouse for complaints
- Utilize consumer expectations to alter payoffs
 - Annual industry convention & product launch
 - Safety advisories & recalls
 - Availability of compatible products

- Manufacturers agreed to TRU deal
- Warehouse club toy sales decreased
 - Pre-agreement growth of 51% per year
 - Overall growth of clubs of over 10% per year

Example: Toys "R" Us

- Sequential timing
 - Threat → Manufacturer's decisions
How will my opponents react?
- Simultaneous timing
 - Manufacturer's decisions
What is my opponent doing right now?
- Information
 - Future profitability of warehouse clubs
What can I infer from the actions of others?
- Agreements
 - Enforceability of contracts & agreements
What happens if someone breaks the agreement?

Defining the Game

CAVEAT

Predict opponents' reaction to your behavior.

BUT

Be sure you understand who your opponents are.

(Do you know *everyone* who may react to your decisions?)



Summary Defining the Game

The strategic environment

- Who are the **players**? (Decision makers)
- What **strategies** are available? (Feasible actions)
- What are the **payoffs**? (Objectives)

Rules of the game

- What is the time-frame for decisions?
- What is the nature of the conflict?
- What is the nature of interaction?
- What information is available?

Origins of Game Theory

- Game Theory was created the Mathematician **John von Neumann**

– von Neumann and Morgenstern's **Theory of Games and Economic Behavior** (1944)

- A mathematician and an economist tried to find a better way to solve certain kinds of economic problems

- Game theory spread from math and economics to sociology, political science, and business



Game Theory

- Game theory – a *theory of interdependent decision making*
- The logic of decision making in social interactions:
 1. *Two or more decision makers (players)*
 2. *Choice of two or more strategies – the outcome of the interaction depends on the strategy choices of all the players*
 3. *Players have well-defined preferences among the possible outcomes; numerical pay-offs can be assigned for all players and outcomes*
- Each player has partial control over the outcome

Games Businesses Play

Standards adoption	coordination
Audits	hunter & hunted
Price wars	prisoner's dilemma
Standards wars	war of attrition
Capacity expansion	commitment
Pollution abatement	free-riding
External financing	hidden information
FCC spectrum	auctions

Understanding Game theory: Some Definitions

Players- Participants of a given game or games.

Rules- Are the guidelines and restrictions of who can do what and when they can do it within a given game or games.

Payoff- is the amount of utility (usually money) a player wins or loses at a specific stage of a game.

Strategy- A strategy defines a set of moves or actions a player will follow in a given game. A strategy must be complete, defining an action in every contingency, including those that may not be attainable in equilibrium

Dominant Strategy - A strategy is dominant if, regardless of what any other players do, the strategy earns a player a larger payoff than any other. Hence, a strategy is dominant if it is always better than any other strategy, regardless of what opponents may do.

Princess

A Gangster and A Movie Actress - Abu Salem and Monica Bedi



- Abu Salem and Monica have been arrested for suspected involvement in "Bombay Blasts" (in 1993), but the police has little evidence.
- The police offered them immunity from prosecution for a confession.
- If both stay silent then both get a short sentence on a minor charge.
- If both confess then both receive a moderate sentence.
- If only one confesses then that criminal goes free and the remaining criminal gets a long sentence.

The Story of Abu and Monica (aka The Prisoner's Dilemma)

		Monica Bedi	
		Do not confess (Cooperate)	Confess (Defect)
Abu Salem	Do not confess (Cooperate)	5 years / 5 years	Life / Parole
	Confess (Defect)	Parole / Life	20 years / 20 years

What's the dilemma?



- Conflict between acting in an **individually rational manner** and in a **collectively rational manner**
- Acting in an **individual** (non-cooperative or competitive) way:
 - *never puts you in a worse position than the other person, irrespective of their behaviour*
- Acting in a **collective (cooperative)** way gives:
 - the best outcome for both individuals
 - but provides the worst outcome if you are the only one acting in this way



Whatsoever ye would that men should do to you, do ye even so to them.



Do not do unto others as you would that they should do unto you. Their tastes may not be the same.



Act only on such a maxim that you can at the same time will to become a universal law.

Prisoners' Dilemma Outcome

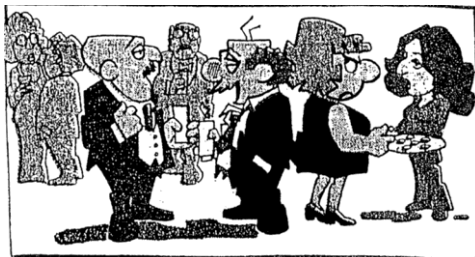
- ❑ **Monica** : Dominant strategy is to confess
- ❑ **Abu Salem** : Dominant strategy is to confess
- ❑ **Dominant strategy equilibrium** – “both confess” (special case of Nash equilibrium).



*If each strategy is a best response given the other strategies in the profile, the profile is a **Nash equilibrium**.*

Perfect Equilibrium!!

“Loretta’s driving because I’m drinking and I’m drinking because she’s driving.”



Dilbert

Interdependence structure

- ❑ The prisoner’s dilemma is only one example of interdependence situation
- ❑ Others might include zero-sum situations
- ❑ The four features of outcome interdependence
 1. degree of dependence
 2. mutuality of dependence
 3. correspondence of outcomes
 4. basis for dependence



A Beautiful Blonde Nash's Coordination Game



A Beautiful Mind

- If everyone competes for the blonde, we block each other and no one gets her.
- So then we all go for her friends.
- But they give us the cold shoulder, because no one likes to be second choice.
- Again, no winner.
- But what if none of us go for the blonde.
- We don't get in each other's way, we don't insult the other girls.
- That's the only way we win.
- That's the only way we all get a girl [for a date!].

Two-by-Two Payoff Table
(Normal Form)

		John	
		Pursue Blonde	Pursue Brunette
Reinhard	Pursue Blonde	(0, 0)	(2, 1)
	Pursue Brunette	(1, 2)	(1, 1)

Blond Date

Understanding Game Theory: Five Assumptions

1. Each decision maker ("PLAYER") has available to him two or more well-specified choices or sequences of choices (called "PLAYS").
2. Every possible combination of plays available to the players leads to a well-defined end-state (win, loss, or draw) that terminates the game.
3. A specified payoff for each player is associated with each end-state (ZERO-SUM game means that the sum of payoffs to all players is zero in each end-state).
4. Each decision maker has perfect knowledge of the game and of his opposition (that is rules of the game and the payoffs of all other players).
5. All decision makers are rational; that is, each player, given two alternatives, will select the one that yields him the greater payoff.

Cigarette Advertising on TV



- All US tobacco companies advertised heavily on TV

- Surgeon General issues official warning
 - Cigarette smoking may be hazardous



- Cigarette companies' reaction
 - Fear of potential liability lawsuits

- Companies strike agreement
 - Carry the warning label and cease TV advertising in exchange for immunity from federal lawsuits.

Strategic Interactions

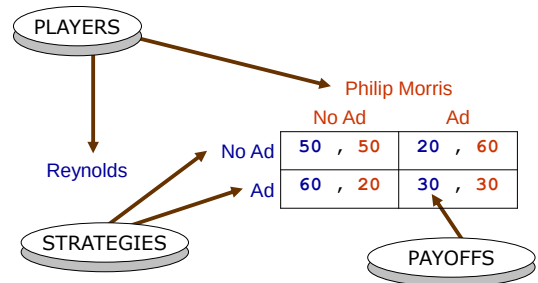
- ❑ Players : Reynolds and Philip Morris
- ❑ Strategies : { Advertise , Do Not Advertise }
- ❑ Payoffs : Companies' Profits

- ❑ Each firm earns \$50 million from its customers
- ❑ Advertising costs a firm \$20 million
- ❑ Advertising captures \$30 million from competitor

How to represent this game?



Normal (Strategic) Form



Normal Form

		Philip Morris	
		No Ad	Ad
Reynolds	No Ad	50 , 50	20 , 60
	Ad	60 , 20	30 , 30

- ❑ Best reply for Reynolds:
 - If Philip Morris advertises : advertise
 - If Philip Morris does not advertise : advertise

- ❑ Regardless of what you think Philip Morris will do

Advertise!

Cigarette Advertising

- ❑ After the 1970 agreement, cigarette advertising decreased by \$63 million
- ❑ Profits rose by \$91 million

Prisoner's Dilemma

- ❑ An equilibrium is NOT necessarily efficient

What if the game is not dominance solvable?

Ransom

No Dominated Strategies

- ❑ Often there are no dominated strategies
- ❑ There may be multiple equilibria
- ❑ **Method:**
Cell-by-cell inspection
- ❑ **Ask:**
Is each player playing the *best response* to the other player?

Retail Store Location

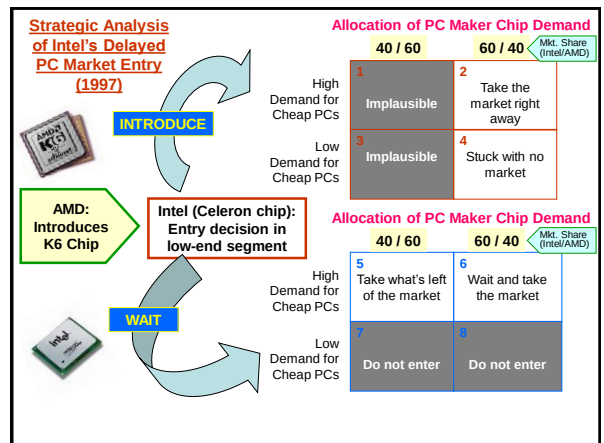
Dept. Stores : **Target** and **Wal-Mart**
 Objective : Choosing a location in a city in USA
 Locations : Uptown, Center City, East Side, West Side

- ❑ If both companies locate in the same part of city, they will split the market there. Unless the market is very large, they will make more money locating in different markets.
- ❑ **Target's** stylish reputation sells well on the wealthy **East Side**.
- ❑ **Wal-Mart's** reputation for value sells well on the middle-class **West Side**.
- ❑ No one does very well Uptown, an area that is a bit down at the heels.
- ❑ Center City draws its customers not from residents but from commuters.

Retail Store Location (payoff table)

		Wal-Mart			
		Uptown	Center City	East Side	West Side
Target	Uptown	30 , 40	50 , 95	55 , 95	55 , 120
	Center City	115 , 40	100 , 100	130 , 85	120 , 95
	East Side	125 , 45	95 , 65	60 , 40	115 , 120
	West Side	105 , 50	75 , 75	95 , 95	35 , 55

Nash Equilib.



Social dilemmas (n - person prisoner's dilemma)

- ❑ **Social dilemma** – a concept in which self-interest and collective interest are at odds
- ❑ **Social traps** – these are social dilemmas in which actions which have a short term attraction to the individual result in a long term cost shared by all (aka, **resource dilemmas**)
- ❑ **Social fences** – social dilemmas in which actions that are unattractive to the individual in the short term are of benefit to the collective in the long term (**public good dilemmas**)
 - Social fences are of special interest as individuals that do not contribute will none the less profit from it
- ❑ **Cooperation** is more likely in resource dilemmas than in public good dilemmas as foregoing immediate benefits is better than paying immediate costs

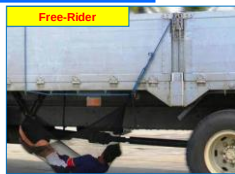
Resolving social dilemmas

- ❑ **Structural solutions** – e.g. limiting the access to a given resource (require an enlightened authority to control)
- ❑ **Group identification** – if a group accesses a shared resource then there is more cooperation if an individual associates themselves strongly with this group
- ❑ **Threat** – a poor solution

Social Dilemmas



Tragedy of Commons



Free-Rider



Chicken (Brinkmanship)



Volunteer's Dilemma

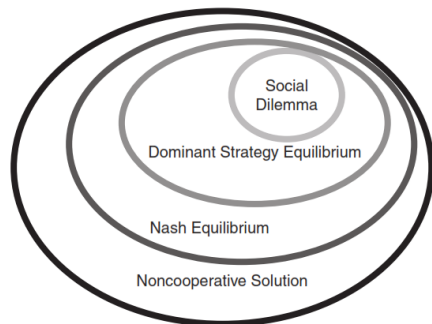
Chennai People

Social Dilemmas

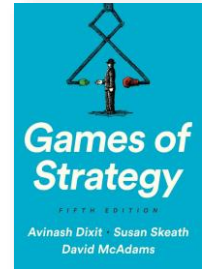
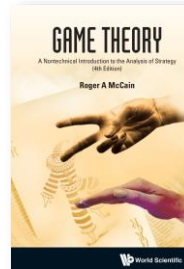


Stag Hunt

Non-cooperative Solution Concepts



A social dilemma is a particular kind of dominant strategy equilibrium, and a dominant strategy equilibrium is a particular kind of a Nash equilibrium, and a Nash equilibrium is a particular kind of non-cooperative solution.



Thank You